PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name	AMAN SAEED
Program / Batch	Networkwalks Cybersecurity Program B084M
Date	12/11/2026
Modules completed	W2-PM1 – Footprinting with Multiple Kali Tools W2-PM5 – Network Scanning with Zenmap
Client / Target	networkwalks.com (authorized scope) Own local LAN network
Permission secured?	Yes – authorized/owned scope only
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery

1. Liability Disclaimer

I performed these activities only on systems and devices that I own or for which I had permission. This report is made for educational and research purposes. The techniques in this report must not be used for unauthorized access. Every person is responsible for using cybersecurity tools in a legal and ethical way.

2. Introduction

This report explains my Week 2 practical activities in the Networkwalks Cybersecurity Program. The first activity was footprinting and reconnaissance of networkwalks.com using different tools in Kali Linux. The second activity was network discovery and scanning of my own authorized local network using Zenmap.

The purpose of these activities was to understand how security professionals collect public information, identify network devices, and document their observations. This report includes the tools used, activities performed, possible security impact, recommendations, and screenshot evidence.

3. Tools Used

Tool	Environment	Purpose
WHOIS	Kali Linux	Collect public domain registration and name-server information.
WhatWeb	Kali Linux	Identify publicly visible website technologies.
Nslookup	Kali Linux	Check DNS information and resolve domain names to IP addresses.
curl -I	Kali Linux	View HTTP response headers.
Wafw00f	Kali Linux	Check for signs of a Web Application Firewall.
DNSRecon	Kali Linux	Collect available DNS record information.
Zenmap (Nmap GUI)	Windows	Discover live hosts on an authorized local network and view topology.
Windows CMD	Windows	Check local IP and network information using ipconfig.

4. Activities Performed

4.1 Footprinting & Reconnaissance (W2-PM1)

Footprinting was performed against networkwalks.com using Kali Linux tools. Each tool was used to collect a different type of publicly available information. The results should always be understood together with the screenshots captured during the practical exercise.

4.1.1 WHOIS

Purpose: To collect public domain registration information such as the registrar, registration dates, and name servers.

4.1.2 WhatWeb
Purpose: To identify technologies that are publicly visible on the website. The practical notes reported WordPress and WP Download Manager information. These details should be checked with the actual screenshot before submission.

4.1.3 Nslookup
Purpose: To resolve the domain name and view available DNS information. The observed address in the practical notes should be verified with the captured command output.

4.1.4 curl -I
Purpose: To inspect HTTP response headers returned by the website. Only information clearly visible in the captured response should be included as evidence.

4.1.5 Wafw00f
Purpose: To check whether a Web Application Firewall can be identified. The practical notes reported ModSecurity, but the final observation should match the screenshot from the practical work.

4.1.6 DNSRecon
Purpose: To collect available DNS records, such as NS, MX, TXT, SPF, SRV, and SOA records.

4.2 Network Scanning with Zenmap (W2-PM5)
The second activity focused on my own authorized local network. Windows Command Prompt was used with ipconfig to check the local IP address and subnet. The subnet was then scanned in Zenmap using a Ping Scan to find active hosts. The Topology view was used to see the discovered devices.

Any IP address, MAC address, or host count in the final report should match the actual results shown in the user's Zenmap screenshots.

5. Risk Analysis / Impact
The observations below are possible information-exposure or network-inventory findings. They are not confirmed vulnerabilities.

#	Observation	Possible Impact	Risk
1	Website technology information is visible	May help an attacker understand the technologies used by the website.	Medium
2	Domain resolves to an IP address	Provides general information about the hosting or network location.	Low
3	HTTP response information is available	May help with technology fingerprinting.	Low
4	WAF technology can be identified	May reveal information about the web security setup.	Low
5	DNS information is available	Can help create a profile of the infrastructure.	Medium
6	Live devices are found on the local network	Unknown devices may need investigation.	Medium

A visible software version, IP address, DNS record, or WAF does not automatically mean that a vulnerability exists. No exploitation or vulnerability validation is claimed in this report.

6. Recommendations

- Review publicly exposed technology information and share only what is necessary.
- Keep the CMS, plugins, server software, and dependencies updated.
- Review HTTP response headers and remove unnecessary information where possible.
- Check DNS records regularly and remove old or unnecessary records.
- Keep the WAF configured, updated, and monitored.
- Perform regular asset discovery on owned or authorized networks.
- Investigate unexpected devices found during network scans.
- Maintain an updated network inventory and topology diagram.
- Always get clear permission and define the scope before scanning or reconnaissance.

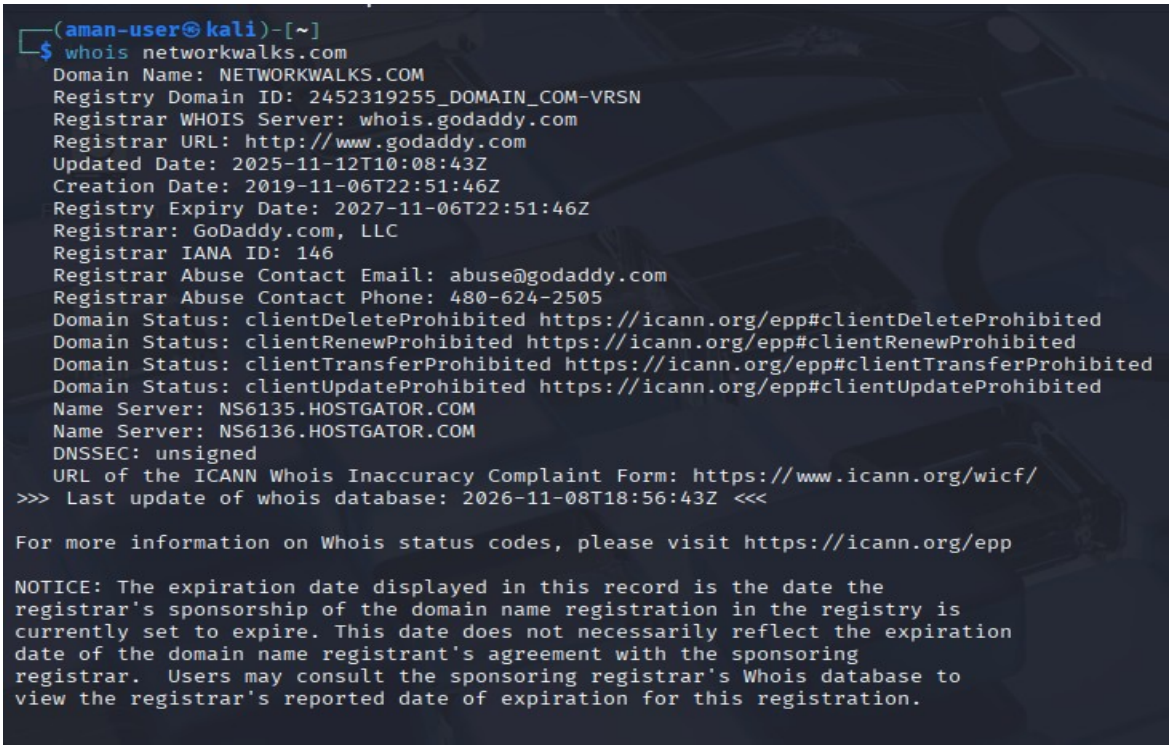
7. Conclusion

During Week 2 of the Networkwalks Cybersecurity Program, I completed practical activities related to footprinting, reconnaissance, and network scanning. I used WHOIS, WhatWeb, Nslookup, curl, Wafw00f, and DNSRecon to understand different types of public domain and website information. I also used ipconfig and Zenmap to review my authorized local network and discover active hosts.

These activities showed the importance of collecting information in a systematic way and documenting the results with screenshots. Information exposure does not always mean that a vulnerability exists. All cybersecurity testing should stay within the approved and authorized scope.

8. Evidence Checklist

WHOIS terminal screenshot

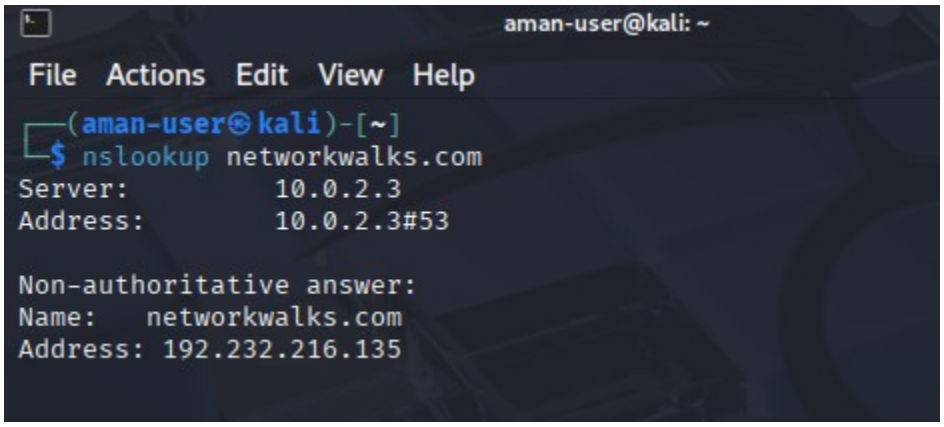


☐ WhatWeb terminal screenshot:

```
┌──(aman-user㉿kali)-[~]
└─$ whatweb networkwalks.com

http://networkwalks.com [301 Moved Permanently] Apache, Cookies[__wpdm_client], Country[UNITED STATES][US], HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216
.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1.4], Cookies[__wpdm_client], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, HTML5, HTTPServer
[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.4,WordPress Download Manager 3.3.76], Open-Graph-Protocol[website],
Script[4684NR-IPIB6amp;pidnVar2=505116amp;prtVar2=76amp;scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Ac
ademy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.4]
```

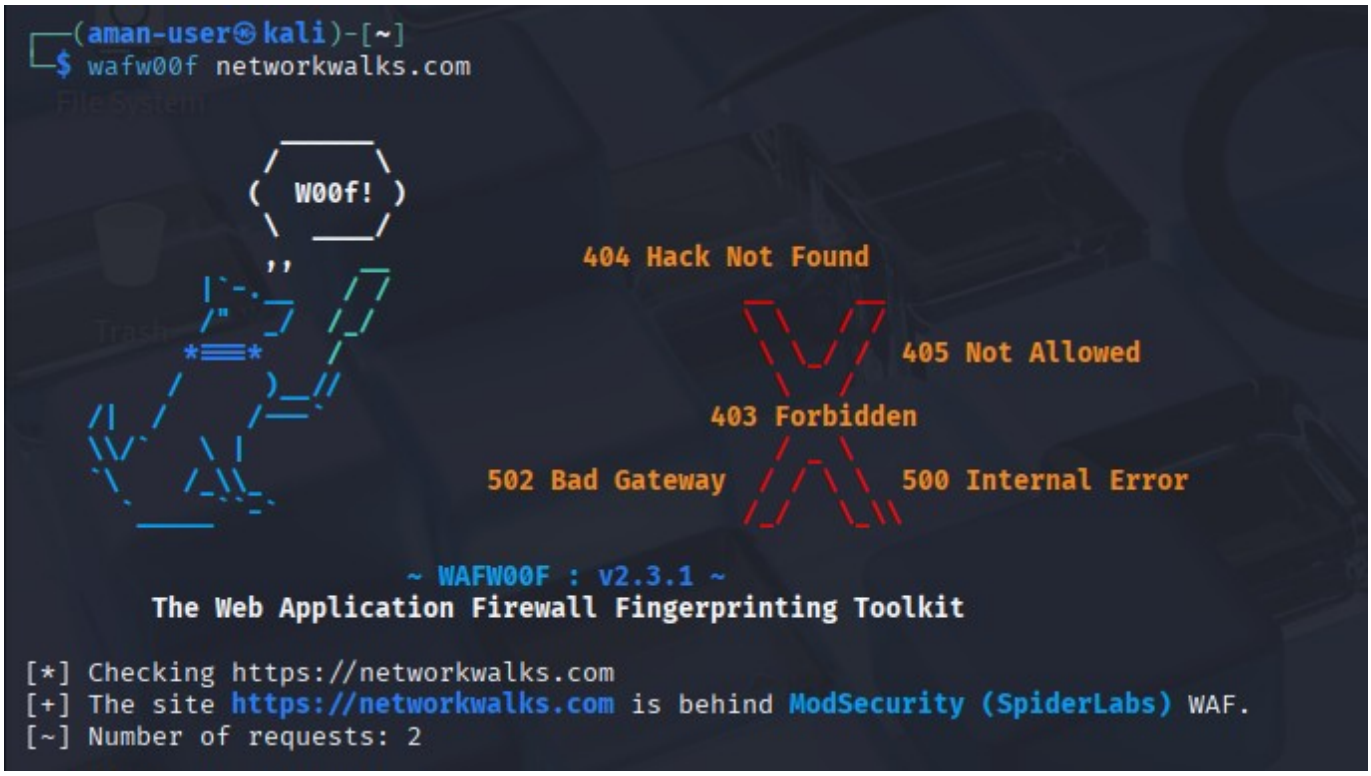
☐ Nslookup terminal screenshot:



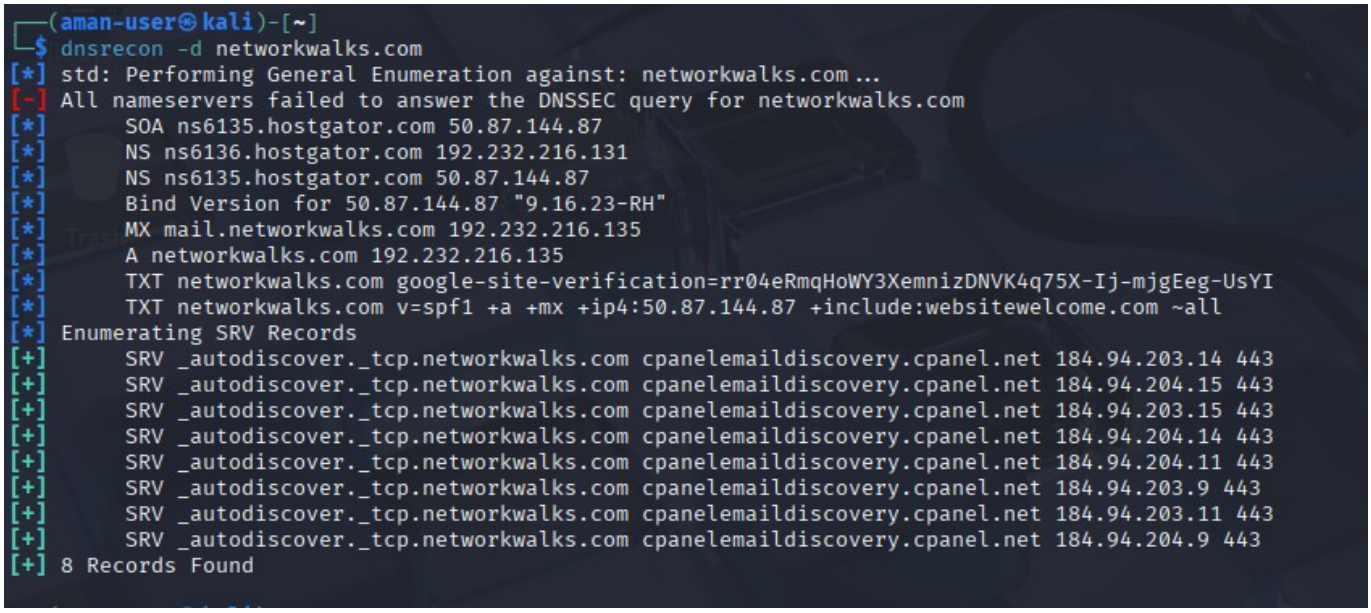
☐ curl -I terminal screenshot:

```
┌──(aman-user㉿kali)-[~]
└─$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.
com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cl
oudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "ht
tps://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "http
s://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https:/
/networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; typ
e="application/json", <https://networkwalks.com/>; rel=shortlink
set-cookie: __wpdm_client=c34f34e01bcde071b7d9d5a4a3e0275b; path=/; domain=ne
tworkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Sun, 08 Nov 2026 20:16:13 GMT
server: Apache
```

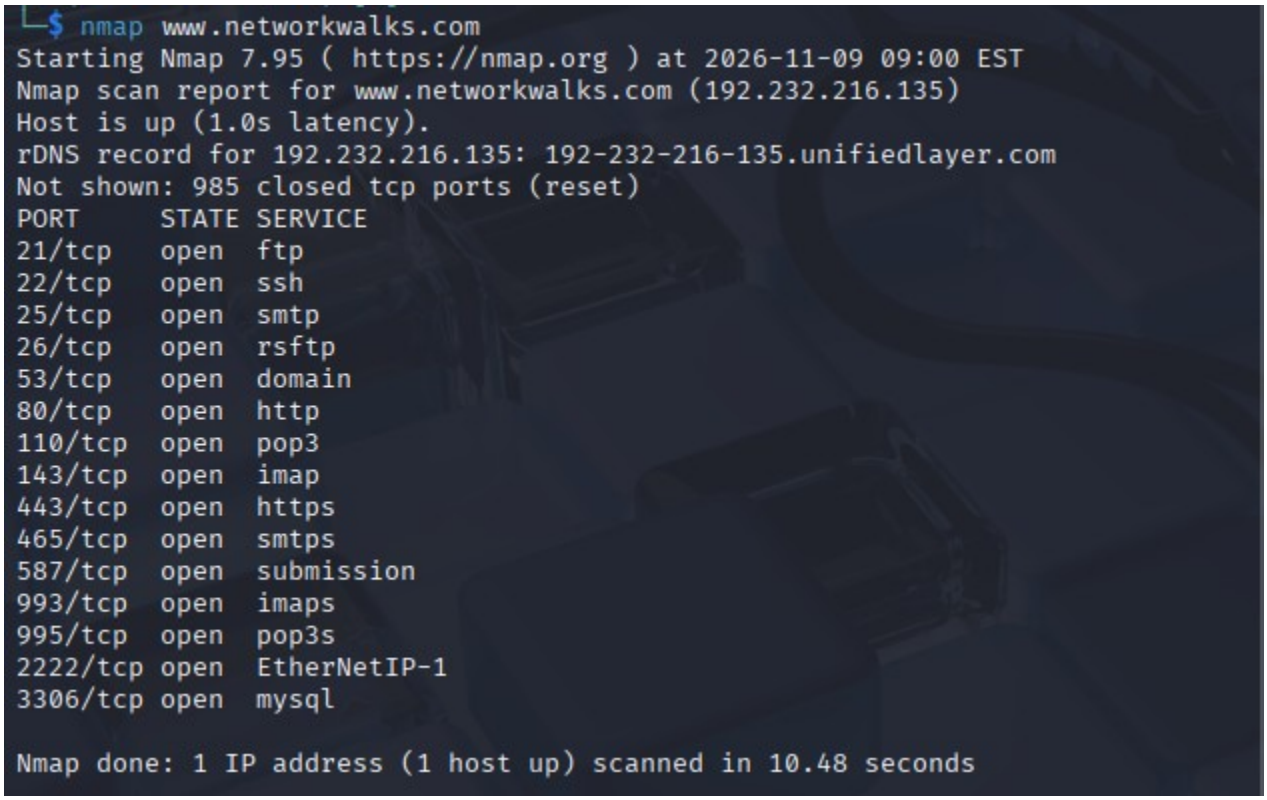
☐ Wafw00f terminal screenshot:



☐ DNSRecon terminal screenshot:



☐ Zenmap host Scan screenshot:



Author / Program Information

Name: Aman Saeed

Program: Cybersecurity Program at Networkwalks | Week: 02

Instructor reference: Waqas Karim, CCIE

GitHub Repository: